David Andrews

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Education

Georgia Institute of Technology, Atlanta, GA: *Ph.D. in Computer Science* — Aug. 2026 – Present
Advisor: Chao Zhang; focusing on AI for scientific research and automating AI research itself

Georgia Institute of Technology, Atlanta, GA: *B.S. in Computer Science* — Aug. 2023 – May 2026
GPA: 4.0/4.0
Courses: DL, ML, Systems for ML, Intro to AI, NLP, Perception and Robotics, Algorithms, OS

Experience

Plato: *Member of Technical Staff (Intern)* — May 2026 – Aug. 2026

- Founded the research and model training arm of Plato, a startup building RL environments for frontier AI labs, expanding the company from environment and task creation into original agent research.
- Built the company's RL training and inference stack from scratch using SGLang and Miles, implementing modern RL algorithms such as SAO for training computer-use agents in Plato's environments.
- Engineered efficient long-horizon, multi-turn computer-use training and inference over image-heavy trajectories, including aggressive caching of multimodal prefixes across turns.

Georgia Tech, Chao Zhang Lab: *Researcher, Artificial Intelligence* — Aug. 2024 – Present

- Leading a team of 3 building a general tool-using agent framework with Docker sandboxes and an MCP interface (browser, terminal, code execution), supporting multi-turn rollouts across math, search, SWE-bench, and puzzle tasks.
- Trained multi-turn LLM agents with GRPO on SWE-bench, SWE-Smith, and SWE-Gym, adding reward-hacking defenses, tool-use checks, and rollout instrumentation to improve training stability.
- Scaled asynchronous PPO/GRPO/RLVR training using AReaL on Ray to maximize throughput under high tool latency, coordinating rollout workers with sandboxes on a Kubernetes cluster across 8-GPU machines.

Georgia Tech, EIC Lab: *Undergraduate Researcher, Hardware Design* — Feb. 2026 – May 2026

- Released **RTL-Universe**, a frontier-difficulty benchmark for LLM agents on Verilog/RTL hardware design, where agents must recreate complete IP designs from only the testbench and verification infrastructure, using Verilator, Yosys, and Icarus Verilog for functional verification.
- Developed the agent harness used to evaluate frontier models on the benchmark's industry-relevant design tasks.

DuckAI: *AI Research Lead* — Aug. 2023 – Present

- Engineered **DuckTrack**, a cross-platform computer-use capture tool adopted by OSWorld and OpenCUA for benchmarking and foundation model training.
- Led development of **PRMBench**'s full-stack annotation platform (React, FastAPI, SQLite), managing a team of annotators and improving labeling speed and inter-annotator agreement for process reward model datasets.
- Built large-scale TTS data pipelines (YouTube and podcast ingestion, upscaling, VAD, diarization) and contributed optimizations to insanely-fast-whisper to accelerate transcription for speech datasets.

Selected Projects

DuckTrack — Nov. 2023 – Jan. 2024

- Extended and maintained precise cross-platform capture of screen, keyboard, and mouse events, synchronizing native input APIs with OBS to achieve millisecond-level sub-frame alignment for computer-use datasets.

Emergent Misalignment — Oct. 2025 – Jan. 2026

- Replicated and investigated emergent misalignment arising from benign fine-tuning of LLMs; findings published on LessWrong.

Skills

Languages: Python, Rust, C, C++, Go, JavaScript, Bash, Verilog
ML: PyTorch, vLLM, SGLang, Miles, Ray, AReaL, veRL, HuggingFace
Systems: Linux, Docker, Kubernetes, SLURM, AWS, FastAPI, SkyPilot, uv, Git

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